

Big Data Analytics (BDA)

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Administrative Introduction

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Interests

Natural Language Processing (NLP)
Computer Vision
Data Analytics
Machine Learning (ML), RNN, Deep Learning (DL)
SotA Transformer Models
Software Development

Administrative: contact details

- Contact Information:
 - ◆ Email: official UCAS Email
 - ◆ Office: Online
 - ◆ Office Hours: N/A

Administrative: reading materials

- Course home page: on LMS, i.e., Google Classroom
 - ♦ Pay attention to course lectures
- Books you may want to read
 - ♦ *List of books that appear in the Syllabus*
- Other Textbooks
 - ♦ *Python for Data Analysis, (2nd edition) by Wes McKinney, O'Reilly, 2017*

Administrative: covered topics

- Overview and motivations
- Create Python Development Environment
 - ◆ Installation & Setup
 - ◆ pip package management tool
- Quick Review:
 - ◆ Python Language Basics, IPython, and Jupyter Notebooks
 - ◆ Built-in Data Structures, Functions and Files
 - Lambda Functions & List comprehension
- Getting Started with pandas
 - ◆ Data Loading, Storage, and File Formats
- Exploratory Data Analysis (EDA)
- Data Cleaning and Preparation
- Plotting and Visualization
- Data Aggregation and Group Operations
 - ◆ Advanced Panda & Pandas-Profiling
- Database Access using Python

Structured

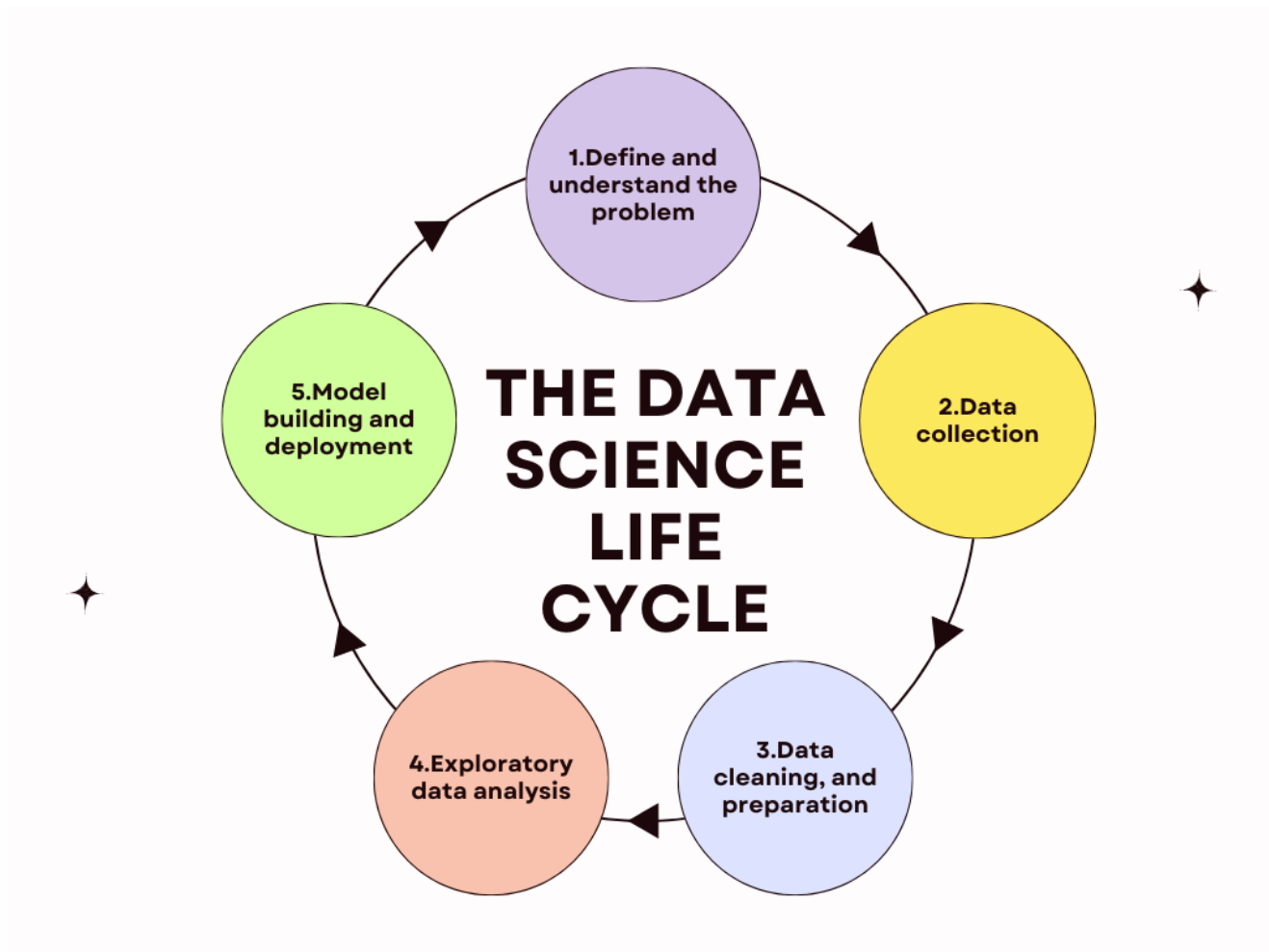
Semi-Structured

Quasi-Structured

Unstructured

Introduction to Data Science <https://sherbold.github.io/intro-to-data-science>

Project Life Cycle



For loop in python

- `for x in range(6):`
 `print(x)`
- `for x in "banana":`
 `print(x)`
- `fruits = ["apple", "banana", "cherry"]`
- `for x in fruits:`
- `if x == "banana":`
- `continue`
- `print(x)`
- `fruits = ["apple", "banana", "cherry"]`
- `for x in fruits:`
- `if x == "banana":`
- `break`
- `print(x)`

List Comprehension

- ```
fruits = ["apple", "banana", "cherry", "kiwi", "mango"]
newlist = []

for x in fruits:
 if "a" in x:
 newlist.append(x)

print(newlist)
```
- ```
fruits = ["apple", "banana", "cherry", "kiwi", "mango"]
newlist = [x for x in fruits if "a" in x]
print(newlist)
```

List Comprehension

- Syntax
 - ♦ `newlist = [expression for item in iterable if condition == True]`
- `newlist = [expression for x in fruits if condition == True]`